| Question | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| 19 | | Proof | M1 | starts process to find point of intersection by substituting, eg $(10 + 2y)^2 + y^2$ (= 20) |
| | | (supported) | M1 | for expanding, eg $4y^2 + 20y + 20y + 100$ (3 out of 4 terms correct) |
| | | | M1 | (dep M2) for 3-term quadratic equation ready for solving, eg $5y^2 + 40y + 80 = 0$ |
| | | | M1 | (dep on previous M1) for method to solve an equation of the form $ay^2 + by + c = 0$, eg by factorising or correct substitution into quadratic formula |
| | | | C1 | fully correct method leading to $y = –4$ or $x = 2$ or $(y + 4)^2 = 0$ or $(x – 2)^2 = 0$ and statement, eg only one point of intersection so the line is a tangent to the circle |